

Area, Perimeter & Volume

Answer sheet · 27 marks total

Section A — Rectangles, triangles & parallelograms

A1. $8 \times 5 = 40 \text{ cm}^2$ (1)

A2. $2 \times (8+5) = 26 \text{ cm}$ (1)

A3. $\frac{1}{2} \times 10 \times 7$ (1) = 35 cm^2 (1)

A4. 13×4 (1) = 52 cm^2 (1)

Section B — Trapeziums

B1. $\frac{1}{2} \times (7+11)$ (1) $\times 6$ (1) = 54 cm^2 (1)

B2. $60 = \frac{1}{2} \times (9+15) \times h$ $60 = 12h$ (1) $h = 60 \div 12$ (1) = 5 cm (1)

Section C — Circles

C1. $\pi \times 7^2$ (1) = 153.9 cm^2 (1)

C2. $\pi \times 18$ (1) = 56.5 cm (1)

C3. Curved part = $\frac{1}{2} \times \pi \times 14 = 22.0 \text{ cm}$ (1) straight edge = 14 cm (1) total = $22.0 + 14 = 36.0 \text{ cm}$ (1)

Section D — Volume

D1. $7 \times 4 \times 5$ (1) = 140 cm^3 (1)

D2. Cross-sectional area = $\frac{1}{2} \times 8 \times 5 = 20 \text{ cm}^2$ (1) volume = 20×12 (1) = 240 cm^3 (1)

D3. $\pi \times 5^2 \times 9$ (1) = $\pi \times 225$ (1) = 706.9 cm^3 (1)

Marking note: award method marks for a correctly set-up calculation or formula substitution even if the final answer contains an arithmetic slip. Accept answers left in terms of π (e.g. $225\pi \text{ cm}^3$) as fully correct unless the question specifically asks for a decimal answer. Missing or incorrect units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.